IIT Bombay Tech Community

Git, GitHub & CI/CD

plus a quick look at Kaggle & Hugging Face

A practical primer for engineering projects —
written alongside a Thermodynamics × AI course project

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1 Why any of this matters

Every project in this document series (the FasterViT thermal-fault detector included) eventually stops being just code on a laptop and starts being something that needs to be **tracked**, **shared**, **tested automatically**, and **built on top of pretrained work**. That is exactly what this primer covers: Git and GitHub for tracking and sharing, CI/CD for automatic testing and deployment, and Kaggle / Hugging Face for the datasets and pretrained models that make an ML project feasible for one person in one semester.

In one line

Git tracks changes, GitHub hosts and shares them, CI/CD automates the checks and deployment, and Kaggle/Hugging Face supply the data and models so you are not starting from zero.

2 Git: Version Control

Git is a distributed version control system — software that records the history of every change made to a project's files, who made it, and why.

2.1 What it actually does

- Takes a **snapshot** of your project every time you `commit`, rather than overwriting the previous version.
- Lets you branch off to try something (a new model head, a new dataset loader) without breaking the working version.
- Lets you **merge** that work back in once it is ready.
- Keeps a full, searchable history — so a bug introduced three weeks ago can be traced to the exact commit that caused it.

2.2 Why it is needed, not just nice to have

- **No more "final_v2_ACTUALLY_final.py".** One history, one source of truth.

- **Safe experimentation.** A branch for fine-tuning experiments can fail without touching the working solver.
- **Reproducibility.** Anyone (including future-you, before an exam) can check out the exact code state that produced a given result.
- **Solo → team ready.** If this becomes a team project later, Git is what prevents overwritten work.

3 GitHub: Hosting, Collaboration & CI/CD

GitHub hosts Git repositories on the web and adds collaboration tooling on top: issues, pull requests, project boards, and — most relevant here — **GitHub Actions**, its built-in CI/CD system.

3.1 Git vs. GitHub, quickly

Git	The tool that tracks versions of your code, runs locally.
GitHub	A website/service that hosts Git repositories online and adds collaboration, review, and automation features.

3.2 CI/CD: what it is

CI (Continuous Integration) automatically builds and tests your code every time you push a change — catching a broken data pipeline or a failing unit test before it reaches the main branch.

CD (Continuous Deployment/Delivery) automatically pushes the validated result somewhere useful once it passes CI — for example, redeploying a Streamlit demo, publishing a model checkpoint, or rebuilding a documentation site.

3.3 Why CI/CD is genuinely needed for this project

Concretely, for the FasterViT fault-detector project

- **Automated testing:** every time the physics-loss calculator or the data-preprocessing script changes, CI runs a small test set automatically — so a bug in the exergy-destruction formula never silently corrupts a week of training runs.
- **Consistent environments:** CI installs dependencies from a clean environment on every run, catching the classic *"it works on my machine"* problem before a professor or recruiter tries to run it.
- **Automatic deployment:** on a successful build, CD can redeploy the Streamlit demo automatically, so the live link on ishaniitb.github.io is always the latest working version — not something you forgot to update.
- **Credibility:** a green CI badge on a GitHub README is a small, concrete signal to anyone reviewing the repository that the project is tested, not just uploaded.

A minimal GitHub Actions workflow for a project like this typically: (1) installs Python dependencies, (2) runs the unit tests on the thermodynamics calculators, (3) checks the model loads and runs one forward pass on a sample image, and (4) if all of that passes on the main branch, redeploys the demo.

4 Kaggle — brief overview

Kaggle is a platform for datasets, competitions, and shareable notebooks with free GPU/TPU access.

- **Datasets:** large, often pre-cleaned public datasets (including thermal-imaging and solar-fault datasets relevant to this project) with clear licensing.
- **Notebooks:** run code directly in-browser with free compute — useful for the fine-tuning step without needing local GPU hardware.
- **Competitions:** a way to benchmark an approach against a public leaderboard, and a recognisable credential on a resume.

5 Hugging Face — brief overview

Hugging Face is the standard hub for pretrained models (including FasterViT-style vision transformers and T5-style language models), plus the `transformers` library used to load and fine-tune them.

- **Model Hub:** thousands of pretrained checkpoints, including vision transformer variants, ready to fine-tune instead of training from scratch.
- **Datasets library:** a standard, versioned way to load and share datasets, similar in spirit to Kaggle but built for direct use in code.
- **Spaces:** free hosting for a live demo (an alternative or complement to a self-hosted Streamlit app) — another good CD target.

6 How it all fits together

Pipeline, end to end

`Kaggle / Hugging Face` (data & pretrained FasterViT) → `local fine-tuning` → `Git` (versioned code & experiments) → `GitHub` (hosted repo, code review) → `GitHub Actions CI/CD` (automated tests, automated redeploy) → `live demo linked from ishaniitb.github.io`

Prepared as a working reference alongside the Engineering Thermodynamics course project.
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